Comparative Analysis of Random Forest Modeling and Clinical Risk Scoring for Diabetes Prediction using the Pima Indians Dataset

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Abstract

Early detection of Type 2 Diabetes is critical for preventing long-term complications, yet traditional clinical screening tools often suffer from limited sensitivity. This study evaluates the predictive performance of machine learning compared to traditional clinical risk scoring for early diabetes detection. Using the Pima Indians Diabetes dataset–a cohort characterized by high disease prevalece–a Random Forest Classifier was compared against a Clinical Baseline model acting as a proxy for American Diabetes Association (ADA) risk guidelines. The methodology involved preprocessing biologically impossible values through median imputation and employing a stratified 80/20 train-test split. The Random Forest model significantly outperformed the manual baseline, achieving an AUC-ROC of 0.80 and a Sensitivity of 0.52, compared to the baseline's 0.61 and 0.30, respectively (see **Table 1**). *These results suggest that integrating ensemble-based classifiers into clinical workflows can reduce diagnostic omissions and enable more precise, population-specific preventative care.*

I. INTRODUCTION

The global prevalence of Type 2 Diabetes necessitates a shift toward more accurate early-warning systems. Traditional clinical risk assessments, while accessible, often rely on simplified point-based heuristics that may fail to capture the complex, non-linear interactions between physiological markers. This research investigates whether AI-based models can provide a more rigorous and sensitive alternative to these traditional methods within the clinical environment. By utilizing the Pima Indians dataset, this project provides a verifiable comparison between modern computational power and established medical screening protocols.

II. RESEARCH OBJECTIVE

The core objective of this study is to answer: How does the predictive performance of AI-based models compare with traditional clinical risk assessment methods for early detection of Type 2 Diabetes in adults?

III. METHODOLOGY

A. Dataset and Preprocessing

This study analyzed 768 health records from the Pima Indians Diabetes Dataset. To ensure data integrity, biologically impossible “zero” values in columns such as Glucose, Blood Pressure, and BMI were replaced with their respective medians. This median imputation helps mitigate the impact of missing data while preserving the overall distribution for model training.

B. Clinical Baseline (ADA Guideline Proxy)

A “Clinical Proxy” risk model was established using standard American Diabetes Association (ADA) risk guidelines. This point-based system assigned risk levels based on age, BMI, and family history to establish a traditional “High Risk” baseline for comparison.

C. Machine Learning Model

A **Random Forest Classifier** was trained to identify patterns within the features. This algorithm was selected for its ability to handle non-linear biological relationships and provide feature importance metrics. The data was partitioned into an 80% training set and a 20% testing set using stratified sampling to maintain class balance (see **Listing 1**).

Listing 1. Python implementation of the data partitioning and 80-20 training/testing split.

```python
if __name__ == "__main__":

  # 1. Load and prep
  print("Loading and preprocessing data...")
  # Local CSV file here
  local_file_path = '/content/diabetes.csv'
  df =  load_and_preprocess_data(local_file_path)
```

```
  # Features and Target
  X = df.drop('Outcome', axis=1)
  y = df['Outcome']

  # Split data (80% train, 20% test)
  X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42, stratify=y)
```

IV. RESULTS AND FINDINGS

The experimental results demonstrate a clear performance gap between the machine learning model and the clinical baseline.

Table 1
Random Forest Model vs. Clinical Baseline Performance Metrics Comparison

Metrics	Clinical Baseline (ADA Proxy)	Model (Random Forest)
Accuracy	0.63	0.75
AUC-ROC	0.61	0.80
Sensitivity	0.30	0.52
Specificity	0.81	0.88

Utilizing **Figure 2**, a Visual Analysis of the ROC curve confirms that the Random Forest model maintains a higher true positive rate across all thresholds compared to the ADA proxy. Notably, feature importance analysis identified **Glucose** levels and **Body Mass Index** (BMI) as the primary drivers of prediction (see **Figure 3**).

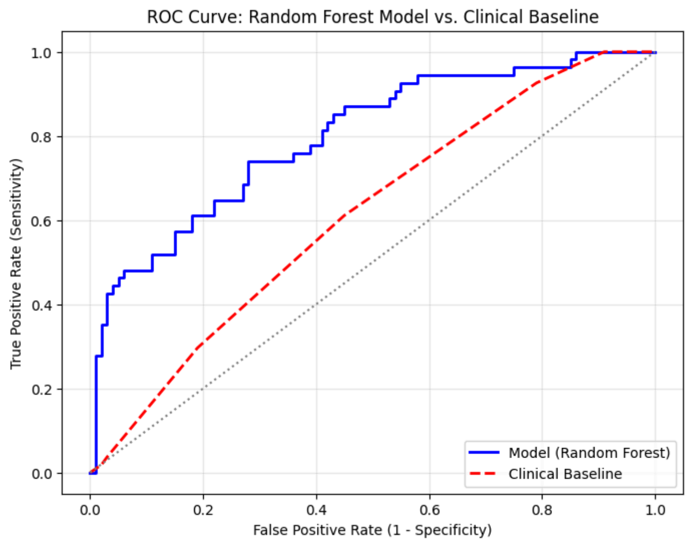

Figure 2. Receiver Operating Characteristic (ROC) curve comparison between the Random Forest model and the Clinical Baseline. The Random Forest model demonstrates superior classification performance, achieving a visibly higher True Positive Rate across all False Positive Rate thresholds compared to the linear clinical baseline. The dotted gray diagonal represents the line of no discrimination (random guessing).

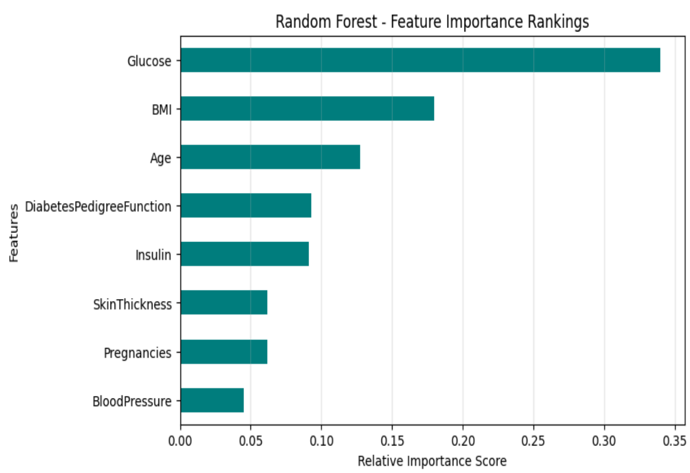

Figure 3. Random Forest feature importance rankings for diabetes prediction. Features are ordered by their relative importance scores derived from mean decrease in impurity (Gini Importance) during model training. **Glucose** exhibits the highest predictive power with a relative score of approximately 0.34, followed by **BMI** (~0.18) and **Age** (~0.13), while **BloodPressure** demonstrates the lowest relative importance.

V. DISCUSSION

The finding that the Random Forest model increased Sensitivity by 22% and AUC-ROC by 0.19 justifies the use of ensemble learning for metabolic risk stratification. While traditional clinical scores are easier to interpret, they may overlook critical factors–such as the Diabetes Pedigree Function–that are vital in high-risk populations. This suggests that AI can significantly reduce diagnostic errors and provide a more nuanced understanding of patient risk.

VI. LIMITATIONS

A. Proxy Constraints

The clinical baseline is a simplified approximation of the ADA guidelines and lacks the subjective nuance or supplementary laboratory data (e.g., HbA1c levels) available to physicians.

B. Demographic Specificity

While the Pima Indians dataset is the holy grail, in terms of diabetes research, results may not be generalizable to other ethnic groups with different genetic and environmental profiles.

C. Missing Data Assumptions

Median imputation assumes that "zero" values were missing at random, though clinical data gaps can sometimes follow informative patterns.

Listing 2. Python implementation of Median imputation to replace impossible "zero" values in certain biological metrics with the appropriate median for that column.

```python
# Values of 0 for certain metrics is impossible. Replace 0s with NaNs,
then impute them with the median of that column.

    cols_to_clean = ['Glucose', 'BloodPressure', 'SkinThickness',
                     'Insulin', 'BMI']
    df[cols_to_clean] = df[cols_to_clean].replace(0, np.nan)
    df.fillna(df.median(), inplace=True)

    return df
```

VII. CONCLUSION

This research concludes that Random Forest modeling provides a superior framework for diabetes risk assessment compared to traditional scoring. The impact of these findings is two-fold: they offer a blueprint for integrating real-time, automated risk stratification into Electronic Health Records (EHRs) and highlight the need for population-specific models to achieve the goals of precision diagnostics.

References

Pradaschnor, N. “diabetes.csv.” Pima-Indians-Diabetes-Dataset, master branch, GitHub, 2018, github.com/npradaschnor/Pima-Indians-Diabetes-Dataset/blob/master/diabetes.csv.

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